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Dr. William R. Young
Editor, JFM
Physical Oceanography Research Division
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Dear Bill,

Enclosed please find two copies of a slightly revised version of the manuscript "Turbulent Diffusion in the Geostrophic Inverse Cascade." We respond to the comments of reviewer B below.

p. 4, end of top paragraph: "... - then the effects of friction become similar to [(a) - this is a typo, I suppose?] those of a barrier..." - I am not sure what this statement means; it should either be better explained or dropped.

This statement was dropped.

p. 13, the discussion of the "halting scale." I still don't understand what this "halting scale" means. If the authors imply termination of the inverse cascade, then this is a wrong statement; the rate of the inverse cascade is ε and termination of the inverse cascade would mean $\varepsilon = 0$ which the authors themselves find unphysical. Equation (5.21) as a definition of the halting scale does not provide more clarity because epsilon remains > 0 for wave numbers smaller than k_r allowing the inverse cascade to penetrate beyond this "halting scale."

The term "halting scale" is accepted by common usage in the literature as the scale at which the efficiency of the cascade begins to decline. Reviewer B is free to challenge this usage in his or her own papers, but we do not find its usage in this context inappropriate.

p. 14, discussion of the combined effect of beta and friction. The authors need to be aware that this effect has been discussed in the paper by Galperin, Sukoriansky and Huang, "Universal n^{-5} spectrum of zonal flows on giant planets," Phys. Fluids, 13, pp. 1545-1548 (2001), to which a proper citation needs to be made.

This reference was added.

p. 17, top paragraph - discussion of the width of zonal jets as related to the steep zonal spectrum. This connection has been explained in the same paper by Galperin *et al.*, (2001).

The paragraph mentioned is primarily concerned with the mixing scale relevant to a passive tracer – comments about the jet scale are made in passing.

p. 17, 3rd paragraph from the top, discussion of the paper by Smith and Waleffe (1999) in which energy transfer was investigated and quartic resonances were invoked. This discussion, almost with the same text and the same references, can be found in a much earlier paper by Chekhlov *et al.*, (1996).

Another citation to Chekhlov was added.

p. 27, discussion of the energy spectra on beta-plane. The authors describe the difference between the $-5/3$ and $-8/3$ spectra in isotropic turbulence and then show zonal -6 spectrum on beta-plane in Fig. 7. To the best knowledge of this reviewer, the zonal -5 spectrum has been found for the zonal modes only in Chekhlov *et al.*, (1996), Smith and Waleffe (1999), and Huang *et al.*, (2001); I am not aware of such spectrum to be found in the entire zonal sector. This means that the -5 spectrum, if it exists, is delta-function like, and, thus, will retain the -5 slope even when calculated along the zonal axis only. The authors need to check this point very carefully.

Perhaps it is a matter of definition of two-dimensional spectra, and Reviewer B may have misunderstood our argument. To be explicit, we meant the following. The total energy E is related to the two-dimensional energy spectrum $\hat{E}(k_x, k_y)$ as

$$E = \int dk_x dk_y \hat{E}(k_x, k_y) = \int dk d\theta k \hat{E}(k, \theta).$$

If we assume that the spectrum is given by

$$\hat{E}(k, \theta) = Ak^{-\alpha} \delta(\theta - \pi/2),$$

similar to the form suggested by Reviewer B, then

$$E = \int dk Ak^{1-\alpha} \int d\theta \delta(\theta - \pi/2) = \int dk Ak^{1-\alpha}.$$

Defining $\mathcal{E}(k)$ (the isotropic spectrum we are usually concerned with) such that

$$E = \int dk \mathcal{E}(k),$$

we have that

$$\mathcal{E}(k) = Ak^{1-\alpha}.$$

Hence if we predict that $\mathcal{E}(k) \propto k^{-5/3}$, then $\alpha = -8/3$. Similarly, if $\mathcal{E}(k) \propto k^{-5}$, then $\alpha = -6$.

Obviously the full spectrum at least contains both parts, and in fact is more complex altogether. But the point is that the $-8/3$ and -6 spectra are just the appropriate dimensional analysis predictions for slices of the two-dimensional spectra along the zonal and meridional axes. We have carefully looked again at our spectra, and there are no additional changes we wish to make in this regard.

Thank you for your time and effort in reviewing our work. We hope that with these changes and comments you will now find the manuscript acceptable for publication in the *Journal of Fluid Mechanics*.

Sincerely,

K. Shafer Smith
Geoffrey K. Vallis